B5.2 Natural selection and evolution

Summary

Variation in the genome and changes in the environment drive the process of natural selection, leading to changes in the characteristics of populations. Evolution accounts for both biodiversity and how organisms are all related to varying degrees. Key individuals have played important roles in the development of our understanding of genetics.

Underlying knowledge and understanding

Learners should appreciate that changes in the environment can leave some individuals, or even some entire species, unable to compete and reproduce leading to extinction.

Common misconceptions

Learners are used to hearing the term evolution in everyday life but it is often used for items that have been designed and gradually improved in order to fit a purpose. They therefore find it difficult to grasp the idea that evolution by natura selection relies on random mutations. Learners also tend to imply that individual change by natural selection. Statements such as 'a moth will change by natural selection in order to become better camouflaged' include both of these commor misconceptions.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers.
All other statements will be assessed in both Foundation and Higher Tier papers.

Topic content			Opportunities to cover:		Practical suggestions
Learning outcomes		To include	Maths	Working scientifically	
B5.2a	state that there is usually extensive genetic variation within a population of a species				
B5.2b	describe the impact of developments in biology on classification systems	natural and artificial classification systems and use of molecular phylogenetics based on DNA sequencing		WS1.1b	
B5.2c	explain how evolution occurs through the natural selection of variants that have given rise to phenotypes best suited to their environment	the concept of mutation			
B5.2d	describe evolution as a change in the inherited characteristics of a population over time, through a process of natural selection, which may result in the formation of new species				
B5.2e	describe the evidence for evolution	fossils and antibiotic resistance in		WS1.1c, WS1.1d,	

B5.2f ☑	describe the work of Darwin and Wallace in the development of the theory of evolution by natural selection and explain the impact of these ideas on modern biology	seedbanks being used as a store of biodiversity		WS1.1a, WS1.1d, WS1.1g, WS1.1h, WS1.3i	